

```
# =====  
# RESOLUTION ALGORITHM  
# Implementation of 5 Propositional Logic Problems  
# =====
```

```
def resolve(clause1, clause2):
```

```
    """
```

```
    Apply the resolution rule to two clauses.
```

```
    Returns a set containing the resolvent(s).
```

```
    """
```

```
    resolvents = set()
```

```
    for literal in clause1:
```

```
        # Find the complementary literal in clause2
```

```
        if literal.startswith("~"):
```

```
            complementary = literal[1:]
```

```
        else:
```

```
            complementary = "~" + literal
```

```
    if complementary in clause2:
```

```
        new_clause = (clause1 - {literal}) | (clause2 - {complementary})
```

```
        resolvents.add(frozenset(new_clause))
```

```
    return resolvents
```

```
def resolution(clauses, goal):
```

```
    """
```

```
    Prove the goal using resolution by contradiction.
```

```
    The negation of the goal is added to the clauses.
```

```
    """
```



```
)
```

```
# Empty clause found
```

```
if len(resolvent) == 0:
```

```
    print("Result: Empty Clause ( $\square$ ) obtained.")
```

```
    print("Therefore, the goal is PROVED.")
```

```
    return True
```

```
new.add(resolvent)
```

```
# If no new clauses can be generated
```

```
if new.issubset(clauses):
```

```
    print("No new clauses can be generated.")
```

```
    print("Therefore, the goal cannot be proved.")
```

```
    return False
```

```
clauses.update(new)
```

```
new.clear()
```

```
def format_clause(clause):
```

```
    """Format a clause for readable output."""
```

```
    if len(clause) == 0:
```

```
        return " $\square$ "
```

```
    return "  $\vee$  ".join(sorted(clause))
```

```
# =====
```

```
# QUESTION 1
```

```
# RAIN AND WET GROUND
```

```
# =====
```

```
print("=" * 70)
```

```
print("QUESTION 1: RAIN AND WET GROUND")
```

```
print("=" * 70)
```

```
print("Goal: Ground is wet")
```

```
clauses_q1 = [
```

```
    {"~R", "W"}, # R -> W
```

```
    {"R"}        # It is raining
```

```
]
```

```
resolution(clauses_q1, "W")
```

```
# =====
```

```
# QUESTION 2
```

```
# STUDENT ASSIGNMENT SUBMISSION
```

```
# =====
```

```
print("\n" + "=" * 70)
```

```
print("QUESTION 2: STUDENT ASSIGNMENT SUBMISSION")
```

```
print("=" * 70)
```

```
print("Goal: Rahul receives internal marks")
```

```
clauses_q2 = [
```

```
    {"~S", "M"}, # S -> M
```

```
    {"S"}        # Rahul submitted assignment
```

```
]
```

```
resolution(clauses_q2, "M")
```

```
# =====
```

```
# QUESTION 3
```

```
# LIBRARY MEMBERSHIP
```

```
# =====
```

```
print("\n" + "=" * 70)
```

```
print("QUESTION 3: LIBRARY MEMBERSHIP")
```

```
print("=" * 70)
```

```
print("Goal: Priya can borrow books")
```

```
clauses_q3 = [
```

```
    {"~L", "B"}, # L -> B
```

```
    {"L"}      # Priya is a library member
```

```
]
```

```
resolution(clauses_q3, "B")
```

```
# =====
```

```
# QUESTION 4
```

```
# PLACEMENT ELIGIBILITY
```

```
# =====
```

```
print("\n" + "=" * 70)
```

```
print("QUESTION 4: PLACEMENT ELIGIBILITY")
```

```
print("=" * 70)
```

```
print("Goal: Arun is eligible for placement")
```

```
clauses_q4 = [  
    {"~A", "P"}, # A -> P  
    {"A"}      # Arun cleared aptitude test  
]
```

```
resolution(clauses_q4, "P")
```

```
# =====
```

```
# QUESTION 5
```

```
# ACCESS CONTROL SYSTEM
```

```
# =====
```

```
print("\n" + "=" * 70)
```

```
print("QUESTION 5: ACCESS CONTROL SYSTEM")
```

```
print("=" * 70)
```

```
print("Goal: User is granted access")
```

```
clauses_q5 = [  
    {"~C", "A"}, # C -> A  
    {"~A", "G"}, # A -> G  
    {"C"}      # Correct password entered  
]
```

```
resolution(clauses_q5, "G")
```

```
# =====
```

```
# FINAL SUMMARY
```

```
# =====
```

```
print("\n" + "=" * 70)
```

```
print("FINAL SUMMARY")
```

```
print("=" * 70)
```

```
print("Q1 - Ground is wet : PROVED")
```

```
print("Q2 - Rahul receives internal marks : PROVED")
```

```
print("Q3 - Priya can borrow books : PROVED")
```

```
print("Q4 - Arun is eligible for placement : PROVED")
```

```
print("Q5 - User is granted access : PROVED")
```

```
print("=" * 70)
```